

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seventh Semester of B. Tech. (IT) Examination
November 2019

IT442 - ADVANCED COMPUTING

Date: 6.11.2019, Wednesday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION-I

Q-1 Answer the following questions.

[07]

1. Which is/are true regarding cloud computing?
(a) It does not provide ubiquitous access
(b) It provides on-demand network access
(c) Resources can be released with no management effort
(d) None of these
2. Amazon Web Services is an example of
(a) SaaS (b) PaaS (c) IaaS (d) None of these
3. The service provider gives the capability to the consumer to deploy onto the cloud in
(a) SaaS (b) PaaS (c) IaaS (d) None of these
4. Email service on cloud is an example of
(a) SaaS (b) PaaS (c) None of these
5. Virtual machine supports migration from one host to another host.
(a) True (b) False
6. Google docs is an example of
(a) SaaS (b) PaaS (c) IaaS
7. An example of hybrid cloud is
(a) Ubuntu Enterprise Cloud – UEC
(b) VMware Cloud Infrastructure Suite
(c) Eucalyptus
(d) VMware vCloud

Q-2 Answer the following questions. (Attempt any Two)

[14]

- [A] Write a short note on Cloud-Adapted Risk Management Framework.
- [B] Explain the following terms in detail with proper examples: Availability, Reliability, Performance, Scalability and Resiliency.

- [C] The overall availability of an IT resource is usually expressed as a percentage of up-time. For example, an IT resource that is always available will have an up-time of 100%. Find out Downtime/Week in seconds for the following table with proper calculations.

Availability	Downtime/Week in seconds
99.5	
99.8	
99.9	
99.95	
99.99	
99.999	
99.9999	

Q-3 Answer the following questions. (Attempt any One)

[14]

- [A] Write a short note on Service Level Agreement Guidelines in detail.
- [B] Write a short note on Microservices with proper real life examples in detail.

SECTION-II

Q-4 Answer the following questions.

[07]

- Which of the following function is responsible for consolidating the results produced by each of the Map() functions/tasks?
(a) Map (b) Reducer (c) Reduce (d) None of these
- When load increases, VM management can be done by
(a) Schedule new VMs to new nodes
(b) Shutdown unused nodes
(c) All of these
(d) None of these
- In Hadoop, which of the following stores metadata?
(a) Data node (b) Name node (c) Client node (d) None of these
- OpenStack provides
(a) Software-as-a-service (b) Platform-as-a-service (c) Infrastructure-as-a-service (d) All of these
- The relationship between the number of processes used and power consumption is
(a) Linear (b) Non-linear
- VM image should as small as possible to reduce network latency.
(a) True (b) False
- Fabrication security attack is an attack on
(a) Confidentiality (b) Availability (c) Integrity (d) Authenticity

Q-5 Answer the following questions. (Attempt Any Two)

[14]

- [A] Compare Virtual Machine Vs Container in detail.
- [B] What are the challenges of cloud computing? Explain with real life examples in detail.
- [C] What are the advantages of using containers?

Q-6 Answer the following questions. (Attempt any Two)

[14]

- [A] Explain Architecture of Grid Computing with proper diagram in detail.
- [B] Why we required virtualization? What are three ways to create virtual servers? Explain in detail with proper examples.
- [C] Compare Traditional Computing vs. Cloud Computing.
